Lecture 24: Trend and Seasonality of Time Series Models and ARCH Models

1 Estimation and Forecasting with ARMA Models

Consider an ARMA model with parameters p and q, given by

$$r_t = \phi_0 + \sum_{i=1}^{p} \phi_i r_{t-i} + a_t - \sum_{i=1}^{q} \theta_i a_{t-i}. \tag{1}$$

ARMA models can be estimated using maximum likelihood. We do not have time to cover it, but you are referred to the book by Brockwell and Davis (2006), "Time Series: Theory and Methods" for the details.

Let us discuss forecasting using an ARMA model. The 1-step ahead forecast r_{T+1} can be easily obtained from the model and is given by

$$\hat{r}_T(1) := \mathbb{E}\left[r_{T+1}\middle|\mathcal{F}_T\right] = \phi_0 + \sum_{i=1}^{p} \phi_i r_{T+1-i} - \sum_{i=1}^{q} \theta_i a_{T+1-i},$$

and the corresponding forecast error is $e_T(1) = r_{T+1} - \hat{r}_T(1) = a_{T+1}$. The variance is $Var\left(e_T(1)\right) = \sigma_a^2$. For the l-steps ahead forecast, we have

$$\hat{r}_T(l) = \mathbb{E}\left[r_{T+l}\middle|\mathcal{F}_T\right] = \phi_0 + \sum_{i=1}^{p} \phi_i \hat{r}_T(l-i) - \sum_{i=1}^{q} \theta_i a_T(l-i),$$

where $\hat{r}_T(l-i) = r_{T+l-i}$ if $l-i \leq 0$, and $a_T(l-i) = 0$ if $l-i > 0$ and $a_T(l-i) = a_{T+l-i}$ if $l-i \leq 0$. Equivalently, we can use the MA representation of an $ARMA(p,q)$ given by

$$r_t = \phi(B)^{-1}\phi_0 + \phi(B)^{-1}\theta(B)a_t = c_0 + \psi(B)a_t = c_0 + a_t + \sum_{i=1}^{\infty} \psi_i a_{t-i}, \tag{2}$$

where $c_0 = \frac{\phi_0}{1-\sum_{i=1}^{p}\phi_i}$ is the common mean μ of the time series. This yields

$$\hat{r}_T(l) = c_0 + \sum_{i=l}^{\infty} \psi_i a_{T+l-i}. \tag{3}$$

We then obtain the forecast error

$$e_T(l) := r_{T+l} - \hat{r}_T(l) = a_{T+l} + \sum_{i=1}^{l-1} \psi_i a_{T+l-i}.$$

Then,

$$Var(e_T(l)) = \mathbb{E}\left[(e_T(l))^2\right] = \sigma_a^2 \left(1 + \sum_{i=1}^{l-1} \psi_i^2\right), \tag{4}$$

which, as expected, is a non-decreasing function of the forecast horizon l. The MA representation of the ARMA model gives a simple proof of mean reversion of a stationary time series. The stationarity implies that ψ_i approaches zero as $i \to \infty$. Therefore, by Eq. (3), we have that $\hat{r}_T(l) \to c_0$ as $l \to \infty$. Because $\hat{r}_T(l)$ is the conditional expectation of r_{T+l} at the forecast origin T, this result says that in the long term the return series is expected to approach its mean c_0, that is, the series is mean reverting. The speed by which $\hat{r}_T(l)$ approaches c_0 determines the speed of mean reversion. Furthermore, using the MA representation given in (2), we have $Var(r_t) = (1 + \sum_{i=1}^{\infty} \psi_i^2)\sigma_a^2$, if the series ψ_i is summable, i.e., if $\sum_{i=1}^{\infty} \psi_i^2 < \infty$. Consequently, comparing equations (2) and (4), we obtain that $Var(e_T(l)) \to Var(r_t)$ as $l \to \infty$.

2 Unit Root Test and Stationarity

We have so far focused on time series which are stationary. This may be a reasonable assumption for returns of securities (e.g. stock returns or bond returns). However, this may not be the case if we consider interest rates, foreign exchange rates, or the price series (as opposed to the return series) of an asset. If we consider a price series, the non-stationarity is typically due to the fact that there is no "fixed level" for the price. Assume, for instance, that the price of the stock follows a geometric Brownian motion. The log-price of the stock follows an arithmetic Brownian motion

$$p_t = \mu + p_{t-1} + a_t,$$

where $\{a_t\}$ is a white noise series. The log-price process p_t is nonstationary. We can rewrite the above equation as $\phi(B)p_t = \mu + a_t$, where the characteristic polynomial $\phi(z) := 1 - z$. This polynomial has a zero within the circle, and in particular it has a *unit root*. This time series fails to satisfy the weak stationarity condition of an AR(1) model.

Using a MA representation, we can write the above series as $p_t = p_0 + \mu t + \sum_{i=1}^{t} a_i$. The term μt is the time trend. Thus, the impact of any past shock a_{t-i} on p_t does not decay over time. Consequently, the series has a strong memory as it remembers all the past shocks. In economics, the shocks are said to have a *permanent effect* on the time series. The strong memory of a unit-root time series can be seen from the sample ACF of the observed series. If the sample ACF does not decay exponentially, it is likely that the time series is non-stationary due to a unit root.

Remark 2.1. *Many people use "nonstationary" interchangeably with "has a unit root". This is one form of nonstationarity, but there are lots of others. For example, if the time series has a structural shift occurring at one point in time, then it would be nonstationary.*

A classical test used for unit roots is the Dickey-Fuller test, which tests the null hypothesis that a unit root is present in an autoregressive model. For example, in an AR(1) model with characteristic polynomial $\phi(z) = 1 - \phi_1 z$, the null hypothesis would be $\phi_1 = 1$. The details of the test are reported in the paper by Dickey and Fuller (1979), and a sketch of the test is given in Section 2.7.5 of the textbook "Linear time series analysis and its applications" by Tsay.

Remark 2.2. *In general, unit root tests have very low power. That is, unit root tests cannot distinguish highly persistent stationary processes from nonstationary processes very well. Also, the*

power of unit root tests diminish as deterministic terms are added to the test regressions. That is, tests that include a constant and trend in the test regression have less power than tests that only include a constant in the test regression. Most recent tests proposed by Elliot, Rothenberg and Stock (1996) and Ng and Perron (2001) have been designed to alleviate these problems.